

DECK 09 · WEEK 5 · MORNING 1

Corporate Governance & WACC

Who owns the firm, who runs it, who finances it — and at what cost

CFA Level I · Vol 3 Learning Modules 1, 2, 3, 6

Roadmap — Corporate Governance, Capital Structure & Cost of Capital

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Organizational forms & ownership

Sole proprietor · partnership · corporation · limited liability · Simon Property Group · public vs private

02

Investors & other stakeholders

Debt vs equity claims · CFA LM2 Example 1 on leverage and ROE · stakeholder map · ESG materiality

03

Corporate governance

Principal-agent · agency-cost arithmetic · dual-class (Magna) · 3 core committees · Theranos / VW / Kobe

04

Capital structure

WACC mechanics · MM I and II · Gerhardt worked twice · Boulder static trade-off · pecking order

05

Cost of capital wrap

CAPM for equity · relevering a comparable beta · after-tax cost of debt · a full WACC end to end

Learning objectives

By the end of this deck you should be able to:

1. Compare the three legal organizational forms of business on five attributes — legal identity, owner-manager relationship, owner liability, taxation, and access to financing.
2. Contrast the financial claims of debt and equity holders and quantify how financial leverage amplifies both shareholder return and risk.
3. Map a corporation's stakeholder groups, explain the principal-agent conflicts between them, and describe the governance mechanisms that mitigate each.
4. Compute a weighted-average cost of capital using market-value weights, an after-tax cost of debt, and a CAPM-based cost of equity.
5. Apply the Modigliani-Miller propositions (with and without taxes), the static trade-off theory, and the pecking order theory to capital-structure decisions.
6. Unlever and relever an equity beta to move between a comparable company's observed risk and the target firm's own capital structure.

SECTION 01

Organizational forms & ownership

Three legal forms of business trade off simplicity for access to capital — sole proprietor, partnership, corporation

The corporate form dominates globally by revenue and assets because it alone offers limited liability, owner-manager separation and unrestricted ownership transfer at the same time.

Attribute	Sole proprietorship	Partnership (GP / LP)	Corporation (public Ltd)
Legal identity	Extension of owner	Extension of partners	Separate legal entity
Owner-manager	Owner operates	Partners operate (GP)	Board hires management
Owner liability	Unlimited	GP unlimited; LP limited	Limited to amount invested
Taxation	Pass-through (owner)	Pass-through (partners)	Company, then dividend tax
Access to financing	Owner + personal debt	Partners + partner loans	Unbounded — public markets

WHY THE CORPORATE FORM WINS ON SCALE

A share is the smallest transferable unit of ownership, and the holder's loss is capped at what they paid for it. Together those two features let a firm pool capital from thousands of investors who want no role in running it. The price is double taxation — profits taxed at the company, then again on distribution — which is why the form suits businesses that retain and reinvest.

¹ Private limited companies (LLC, SARL, GmbH, K.K.) keep limited liability but restrict ownership and are usually taxed pass-through.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, §2 "Organizational Forms of Businesses", Exhibit 6, pp. 6–13.

Corporations combine limited liability with owner-manager separation — the combination that unlocks deep capital markets

Shareholders → elect → board of directors → appoints → management. Three layers, each meant to act for the layer that put it there.

THE FIVE FEATURES CFA ACTUALLY LISTS

1. Legal identity — a separate person in law: contracts, hires, sues, borrows, pays tax
2. Owner-manager separation — shareholders elect the board; the board appoints the CEO
3. Owner liability — limited: a shareholder can lose the purchase price and no more
4. External financing — buying a share is the only requirement to become an owner
5. Taxation — profits taxed at the company, then again on distribution

Ownership is divided into small, standardised, freely transferable shares. That is the mechanism which makes (3) and (4) work together.

SIMON PROPERTY GROUP — A HYBRID STRUCTURE

Simon holds over USD 33bn of retail real estate in two layers:

- Simon Property Group L.P., a limited partnership, owns the assets outright.
- Simon Property Group Inc., a corporation, is the general partner with an 87% interest — and its only asset is that partnership stake.

The GP carries unlimited liability, but the GP is itself a company whose shareholders do not. The ultimate owners are shielded while profit and loss are still shared with around 200 limited partners. The founding family holds 8% of Inc.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, §3 "Key Features of Corporate Issuers", pp. 14–19. Simon Property Group is Example 1, pp. 10–11.

Public companies trade liquidity and price transparency for disclosure, listing costs and the scrutiny of outside shareholders

Going public — or going private — is a trade between access to capital and freedom from continuous public scrutiny. The legal form itself does not change.

Dimension	Public (listed)	Private (unlisted)
Liquidity	Continuous exchange trading	Illiquid – negotiated blocks
Price transparency	Real-time price, regulated news flow	Negotiated, stale, often opaque
Disclosure	Annual audited, interim reviewed	Bilateral, often confidential
Access to capital	Broad – public debt & equity markets	Narrow – relationship-based
Shareholder base	Thousands, retail + institutional	Founders, PE, family, strategics
Free float	Shares not held by insiders trade	None – accredited investors only
Governance scrutiny	Activists, proxy fights, say-on-pay	Largely insulated from markets

MOVING BETWEEN THE TWO

Private → public, three routes (CFA Exhibit 16): IPO — underwritten, new shares issued, cash raised · direct listing — no underwriter, no new shares, existing holders sell · acquisition — either by a listed company or by a SPAC, the "blank cheque" shell that has replaced the 1990s reverse merger. Public → private: investors buy in every listed share at a premium, usually with debt, then delist.

Source: CFA Institute, Curriculum Vol 3, Learning Module 1, §4 "Publicly vs. Privately Owned Corporate Issuers", Exhibits 10 and 16, pp. 20–32.

SECTION 02

Investors & other stakeholders

Debt is a fixed, priority, finite claim; equity is a residual, discretionary, permanent claim — the two are economically asymmetric

One pool of cash flow, two claims on it. Debtholders are paid in full or not at all; shareholders get whatever is left over.

Attribute	Debt	Equity
Tenor	Finite – 3m to 30y	Permanent, no maturity
Claim type	Fixed and senior	Residual, discretionary
Payment	Contractual interest + principal	Dividend only if declared
Return potential	Capped at the promised rate	Unlimited in theory
Liquidation priority	Paid before any equity claim	Last in line – residual
Tax treatment	Interest deductible for issuer	Dividends paid post-tax
Voting rights	None	Yes – board, M&A, charter
Risk to the issuer	Higher – must be serviced	Lower – no obligation
Risk to the investor	Lower – priority, fixed	Higher – residual claim

THE SAME CASH FLOWS, MIRRORED RISK

Issuer view: debt is riskier — it must be serviced. Investor view: equity is riskier — it is residual.

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, §2 "Financial Claims of Lenders and Shareholders", pp. 39–47.

Leverage amplifies both return and risk — the same mechanism does both, and CFA Example 1 proves it in three lines

Same operating business, two capital structures, three revenue scenarios. Leverage turns a 30% return on assets into a 60% ROE — and a 20% revenue drop into a loss.

Scenario	Equity-financed (E 100)	Debt-financed (D 75 / E 25, int. 15)	ROE gap
Base — Rev 100, OpEx 70	NI 30 / E 100 = 30%	NI 30-15 = 15 / E 25 = 60%	+30pp
Upside — Rev 120 (+20%)	NI 50 / E 100 = 50%	NI 50-15 = 35 / E 25 = 140%	+90pp
Downside — Rev 80 (-20%)	NI 10 / E 100 = 10%	NI 10-15 = -5 / E 25 = -20%	-30pp

WHY IT WORKS — AND WHERE IT BREAKS

Leverage pays only while the return on assets beats the cost of debt: here return on assets is $30/100 = 30\%$ against an interest rate of $15/75 = 20\%$. On the upside the same gearing turns a 50% all-equity return into 140% on equity. On the downside a still-profitable business — assets earning 10% — produces a net loss of 5 on 25 of equity: a -20% ROE that removes a fifth of book equity in a single period and puts the next interest payment in question. Leverage does not raise the average return. It widens the distribution around it.

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, Example 1 "Equity- versus Debt-Based Balance Sheet Financing", pp. 42–43. Income taxes ignored, as in the source.

Eight stakeholder groups each bring a distinct set of claims, incentives and conflict potential with the corporation

Shareholder theory counts only the residual claimants. Stakeholder theory widens the lens to everyone with a vested interest — and ESG analysis follows directly from it.

Stakeholder	What they provide	What they want from the firm
Shareholders	Equity capital	Residual return, votes, appreciation
Debtholders / creditors	Debt capital	Interest, principal, covenant cover
Board of directors	Stewardship and oversight	Fiduciary duty, fees, reputation
Managers	Human capital and strategy	Pay, status, tenure, career options
Employees	Labour and skills	Fair pay, safety, job security
Customers	Revenue	Value, service, safety, supply
Suppliers	Inputs and trade credit	Timely payment, long relationship
Governments / society	Rule of law, infrastructure	Tax, compliance, externalities

SHAREHOLDER THEORY VS STAKEHOLDER THEORY

Shareholder theory counts other groups only where they affect shareholder value; stakeholder theory makes them objectives in their own right.

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, §3 "Corporate Stakeholders and Governance", Exhibit 6, pp. 48–53.

ESG factors are increasingly quantifiable and material — analysts push them into cash-flow forecasts and into the discount rate

Equity is hit first and hardest because it is the residual claim. Debt follows only if payment capacity is impaired — yet all three companies below were downgraded.

ENVIRONMENTAL

Climate — physical and transition risk · pollution and waste · resource and land use · biodiversity · stranded assets

Most material in utilities, oil and gas, mining, autos, airlines, food.

SOCIAL

Human capital · data privacy and security · product safety · community relations · labour standards

Most material in tech, healthcare, consumer, pharma.

GOVERNANCE

Board composition · audit committee · executive pay · voting rights · bribery and corruption

Material for every issuer, and the most quantifiable of the three — which is why analysts priced it first.

Case in the curriculum	Factor	What it cost
Vale — Brumadinho dam, 2019	Environmental	270 deaths, USD 7bn settlement, rating cut to junk
Equifax — data breach, 2017	Social	140m records stolen, hundreds of USD m in costs
Siemens — bribery, to 2008	Governance	EUR 3bn+ fines, both board chairmen resigned

Source: CFA Institute, Curriculum Vol 3, Learning Module 2, §4 "Corporate ESG Considerations", Exhibit 7 and case boxes, pp. 54–61.

SECTION 03

Corporate governance

The corporation is a chain of principal-agent relationships — every link opens a conflict of interest and an agency cost

Shareholders (principal) elect directors (agent); directors (now principal) hire managers (agent). At every link the agent knows more than the principal.

Agency cost = direct monitoring cost + indirect forgone value

- 1 Insufficient effort**
Shirking: too little monitoring, hard decisions ducked, attention split across outside boards and side projects.
- 2 Inappropriate risk appetite**
All-cash pay makes managers too cautious; options-heavy pay makes them reckless — the upside is theirs, not the downside.
- 3 Empire building**
Pay and status track company size, so growth becomes the objective — acquisitions that enlarge the firm and destroy value.
- 4 Entrenchment**
Job protection: poison pills, staggered boards, copying peers, deals complex enough that only the incumbent can run them.
- 5 Self-dealing**
Excessive perquisites — jets, clubs, personal security — and related-party deals. Owners pay, the manager consumes.

These five are the manager ↔ shareholder conflicts. Two further axes follow: controlling ↔ minority shareholders (next slide) and shareholders ↔ creditors.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, §2 "Stakeholder Conflicts and Management", Exhibit 1, pp. 70–75.

Pricing one perk turns soft governance talk into a valuation number — and shows why boards fix it for free

Perks that look immaterial on a P&L capitalise into large numbers. A governance problem is a valuation problem in disguise.

ATLAS INDUSTRIAL — PRICING A SELF-DEALING PERK

The CEO of Atlas uses a company-leased jet for personal travel and the board does not require reimbursement. This is self-dealing — excessive perquisites — not empire building. Atlas' cost of equity is 10% and its marginal tax rate 25%. What is the perk worth to shareholders?

1. Annual cost: USD 3.2m lease + USD 0.8m fuel and crew = USD 4.0m pre-tax
2. The company deducts it, so owners bear $4.0 \times (1 - 0.25) = \text{USD } 3.0\text{m}$ a year after tax
3. Capitalise the perpetuity at the cost of equity: $\text{PV} = 3.0 / 0.10 = \text{USD } 30\text{m}$
4. On 15m shares that is USD 2.00 per share transferred from owners to one executive
5. Fix: charge personal legs at charter rates — 2.5m pre-tax is 1.875m after tax, PV 18.75m, USD 1.25 a share

PV of the agency cost is about USD 30m; the fix recovers USD 18.8m of it.

Source: Illustrative company ("Atlas Industrial"), built on the agency-cost framework in CFA Vol 3, Learning Module 3, pp. 70–71. Not a curriculum example.

Concentrated ownership and dual-class shares shift power to insiders — and move the conflict from manager-versus-owner to majority-versus-minority

CFA Institute has long advocated against dual-class structures: they let one group override the will of the majority. Where they are legal, it demands clear disclosure.

DISPERSED OWNERSHIP

No single holder controls. The live conflict is shareholder ↔ manager. An activist can force change through a proxy contest.

CONCENTRATED OWNERSHIP

A founder, family, PE house or the state holds a controlling stake. The conflict shifts to controlling ↔ minority. It also buffers the firm from a hostile bid.

DUAL-CLASS SHARES

Class A, listed, one vote. Class B, insiders, many votes. A small economic stake controls the company — and once adopted the structure is very hard to unwind.

MAGNA INTERNATIONAL — HOW 3% OF THE SHARES CONTROLLED 75% OF THE VOTES

Class A carried one vote, Class B carried 500. Frank Stronach and his family held about 3% of Magna's shares and 75% of the votes, while drawing consulting fees, salaries, bonuses and options through a weak share price.

Shareholders voted to collapse to one class — and paid CAD 870m over five years.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, "Controlling and Minority Shareholder Relationships", pp. 72–73.

Governance mechanisms fall into seven categories — each one built for a specific principal-agent conflict

Reporting shrinks information asymmetry; shareholder, creditor and employee mechanisms police the contract; board and government mechanisms set the rules of the game.

Category	Primary tools	Conflict it targets
Reporting & transparency	Audited accounts, proxy, disclosure, IR	Information asymmetry
Shareholder mechanisms	AGM, EGM, proxy contest, say-on-pay, activism	Manager ↔ shareholder
Creditor mechanisms	Indenture, covenants, trustee, ad hoc cttee	Shareholder ↔ creditor
Board & management	Audit / Nom / Comp committees, LTIPs	Manager ↔ shareholder
Employee mechanisms	Labour law, unions, contracts, ESOPs	Firm ↔ employees
Customer & supplier	Commercial contracts, warranties, recourse	Firm ↔ counterparties
Government / regulator	Securities law, governance codes, SOX	Issuer ↔ society

COMPLY OR EXPLAIN

Most jurisdictions run a "comply or explain" code — adopt the recommended practice, or publicly justify the departure. Japan, for instance, makes a company with no outside directors say why. The US has no national code at all: it relies on the Securities Act and Sarbanes-Oxley, SEC enforcement, exchange listing rules, and Delaware corporate law.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, §3 "Corporate Governance Mechanisms", pp. 76–83.

Three core board committees — Audit, Nominating/Governance, Compensation — put independence exactly where it is needed

Committees do the first-level work and report up; the board cannot delegate its ultimate responsibility. Best practice on all three: composed of independent directors.

AUDIT COMMITTEE

Mandate: integrity of financial reporting, accounting policy, internal audit, appointment and fee of the external auditor, whistleblower channel — sometimes IT security too.

Best practice:

- Solely independent directors
- At least one with accounting or financial expertise
- Meets the auditor without management present

Red flag: an executive sitting on it.

NOMINATING / GOVERNANCE

Mandate: director selection and the board election process, board composition and balance, the governance code, the board charter, the code of ethics, the conflicts-of-interest policy.

Best practice:

- Independent members
- Explicit skills matrix
- Annual board evaluation

Red flag: the CEO picking the directors.

COMPENSATION

Mandate: pay policy for directors and key executives, performance criteria, equity grants, the say-on-pay vote.

Best practice:

- All members independent — management must not assess itself
- Shares rather than options, vesting over years
- Clawback provision

Red flag: a one-year cash bonus on a single metric.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, "Board and Management Mechanisms", Exhibit 2, pp. 80–82.

Weak governance raises the cost of capital – Theranos, Volkswagen and Kobe Steel each priced it differently

Three failures, three transmission channels: a total equity loss, a legal and reputational bill, and a step change in the cost of debt.

THERANOS (US, 2003-18)

A celebrated board with almost no medical-technology expertise, which dismissed whistleblowers and never hired an independent expert.

- Raised over USD 700m; valued at USD 10bn in 2014
- Worth zero by 2018; operations ceased
- Founder convicted on four counts of fraud, 2022

Root cause: inadequate board composition and oversight.

VOLKSWAGEN (DE, DIESELGATE)

Illegal "defeat device" software fitted to as many as 11m VW and Audi cars, denied for a year before the 2015 admission.

- Share price fell over a third within days
- EUR 32bn of recalls, fines and legal costs
- CEO resigned, then indicted in the US and Germany

Root cause: weak board independence and no oversight of the teams that installed it.

KOBE STEEL (JP, 2017-18)

Strength and durability data falsified for aluminium, copper, steel and iron ore powder; admitted in March 2018.

- Stock to a five-year low
- Cost of debt jumped to record levels
- Remedy: independent chair, a third of the board independent, three independents on a five-member audit committee

Root cause: a culture putting orders and deadlines ahead of quality, with no quality-control oversight.

¹ Evidence cited by CFA: Gompers-Ishii-Metrick (2003) on governance and valuation; Bhojraj-Sengupta (2003) on bond ratings and yields.

Source: CFA Institute, Curriculum Vol 3, Learning Module 3, §4 case boxes, pp. 86–88; Kobe Steel is Practice Problems 1–5, pp. 92–94.

SECTION 04

Capital structure

WACC blends the after-tax cost of debt with the cost of equity – the single most important discount rate in corporate finance

WACC is three things at once: the r in an NPV, the hurdle rate for an IRR, and the return investors require weighted by how the firm is financed.

$$\text{WACC} = w_d \cdot r_d \cdot (1 - t) + w_e \cdot r_e$$

where: w_d , w_e = market-value weights, or management's target (usually set on book values); r_d = pre-tax cost of debt; r_e = cost of equity; t = marginal tax rate.

CFA KNOWLEDGE CHECK – ABC CORPORATION

ABC is financed 40% debt / 60% equity and intends to keep those proportions. Debt investors require 4% before tax, equity investors 10%. The marginal tax rate is 23% and interest is deductible.

1. After-tax cost of debt: $4\% \times (1 - 0.23) = 3.08\%$
2. Debt contribution: $0.40 \times 3.08\% = 1.23\%$
3. Equity contribution: $0.60 \times 10\% = 6.00\%$
4. WACC = $1.23\% + 6.00\% = 7.23\%$

WACC = 7.23% – the rate for a project of average firm risk.

¹ CFA Equation 1 states WACC without the tax term; the $(1 - t)$ factor applies to the cost of debt wherever interest is deductible.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §2 "The Cost of Capital", Equation 1 and Knowledge Check: Computing WACC, pp. 178–182.

Book weights and market weights give different answers – and for a cost of capital it is the market weights that count

Market values are what investors could realise today, so they measure opportunity cost. Book values are what was paid, years ago.

Meridian Corp – capital stack	Book value	Market value	BV weight	MV weight
Long-term debt (5.0% coupon, 7y, YTM 4.0%)	400	420	33%	26%
Common equity (60m shares × USD 20)	800	1,200	67%	74%
Total capital	1,200	1,620	100%	100%

ASSEMBLE THE WACC – MARGINAL TAX RATE 25%

Pre-tax cost of debt = YTM = 4.0%. Cost of equity from CAPM = 9.5%. Use market-value weights.

1. After-tax cost of debt: $4.0\% \times (1 - 0.25) = 3.00\%$
2. Debt contribution: $0.26 \times 3.00\% = 0.78\%$
3. Equity contribution: $0.74 \times 9.50\% = 7.03\%$
4. WACC = $0.78\% + 7.03\% = 7.81\%$ on market weights, against 7.33% on book weights

WACC = 7.81%. Book weights flatter the firm by 48bp.

¹ Market weights for the WACC; but management targets, covenant tests and rating-agency ratios are all stated on book values.

Source: Illustrative company ("Meridian Corp") built on CFA Vol 3, Learning Module 6, §2, pp. 178–182. The curriculum works the same point in Question Set 3–4, pp. 181–182.

Business model and life-cycle stage set the mix — startups, growth firms and mature firms are financed in visibly different ways

Business risk falls and free cash flow stabilises as a firm matures, so lenders respond with bigger cheques, tighter spreads and looser terms — in that order.

Life-cycle feature	Startup	Growth	Mature
Revenue growth	Initial	Rising	Slowing
Free cash flow	Negative	Rising, often still <0	Stable, positive
Business risk	High	Medium	Low
Debt availability	Little or none	Rising	High
Typical debt type	Convertible, leases	Secured	Unsecured
Typical equity source	Founders, VC	Growth equity, PE	Retained earnings
Illustrative firm	Biotech pre-approval	Tesla circa 2015	Coca-Cola, P&G

CAPITAL INTENSITY CUTS ACROSS EVERY STAGE

Uber and Airbnb run capital-light networks; utilities, semiconductors and airlines are capital-intensive. Intensity decides HOW MUCH capital is needed; the life-cycle stage decides WHAT TYPE — and CFA Exhibit 2 maps that stage straight onto convertible, then secured, then unsecured.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §3 "Factors Affecting Capital Structure", Exhibit 2, pp. 185–186.

MM Proposition I without taxes: in a frictionless world capital structure is irrelevant — value is the PV of the cash flows, full stop

Modigliani and Miller, 1958 — the starting point of modern corporate finance, and the benchmark against which every real-world deviation gets measured.

$$V_L = V_U \quad - \quad \text{and WACC is unaffected by the mix}$$

where: V_L = value of the levered firm; V_U = value of the same firm all-equity. Holds only under the five MM assumptions below.

MM assumption (CFA Exhibit 3)	What it rules out — and the reality
1. Homogeneous expectations	Investors disagree about cash flows
2. Perfect capital markets	Interest is deductible; distress is costly
3. Risk-free borrow and lend	Individuals pay more than corporates do
4. No agency costs	Managers do not always act for owners
5. Financing / investment separate	They interact — signalling, covenants

HOMEMADE LEVERAGE — AND THE ASSUMPTION EVERYONE MISSTATES

An investor who wants 70% leverage at a 50%-levered firm simply borrows on personal account, so arbitrage forces $V_L = V_U$. And note: MM did not assume bankruptcy away. They assumed it happens at zero cost.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §4 "Capital Structure Irrelevance (MM Proposition I without Taxes)", Exhibit 3, pp. 193–195.

MM Proposition II without taxes: more debt raises the cost of equity by exactly enough to offset the cheaper debt, so WACC does not move

Debt is cheaper because it is senior. Adding it makes the remaining equity riskier by precisely the amount that cancels the saving.

$$r_e = r_0 + (r_0 - r_d) \cdot D / E$$

where: r_e = cost of levered equity; r_0 = cost of equity with no debt, which equals WACC at $D = 0$; r_d = cost of debt; D and E at market value.

CFA EXAMPLE 2 — GERHARDT CORP, ALL-EQUITY THEN A DEBT-FUNDED BUYBACK

Gerhardt has a perpetual cash flow of EUR 5,000 and a cost of equity of 10%, which is also its WACC. No taxes, so value = $5,000 / 0.10$ = EUR 50,000. It now issues EUR 15,000 of 5% debt and buys back the same amount of equity.

1. After the buyback: $D = 15,000$, $E = 35,000$, so $D/E = 0.4286$
2. New cost of equity: $r_e = 10\% + (10\% - 5\%) \times (15,000 / 35,000) = 12.143\%$
3. New WACC: $(15/50) \times 5\% + (35/50) \times 12.143\% = 1.50\% + 8.50\% = 10.00\%$
4. Cross-check: $750/0.05 + 4,250/0.12143 = \text{EUR } 50,000$. Value unchanged.

WACC = 10% before and after. Cheaper debt, exactly offset by dearer equity.

¹ CFA Exhibit 4 plots it: r_e rises linearly in D/E from an intercept of r_0 with slope $(r_0 - r_d)$, while WACC stays flat.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §4, Equation 4 and Example 2 "Gerhardt Corporation Cost of Equity", pp. 195–197.

With corporate taxes the shield makes debt value-creating: $V_L = V_U + tD$, the cost of equity rises more slowly, and WACC falls

Once profits are taxed, every euro of deductible interest is a euro the firm keeps. Capital structure starts to matter — and at the extreme this says finance with 100% debt.

$$\text{I: } V_L = V_U + t \cdot D$$

$$\text{II: } r_e = r_0 + (r_0 - r_d)(1 - t) \cdot D/E$$

where: t = marginal corporate tax rate; $t \cdot D$ = PV of the interest tax shield on perpetual debt. Only the $(1 - t)$ term separates II here from II without taxes.

CFA EXAMPLE 3 — THE SAME GERHARDT, NOW TAXED AT 25%

Same EUR 5,000 pre-tax cash flow, same 10% unlevered cost of equity, same EUR 15,000 of 5% debt. After-tax cash flow is 3,750, so $V_U = 3,750 / 0.10 = \text{EUR } 37,500$.

1. PV of the tax shield: $t \cdot D = 0.25 \times 15,000 = \text{EUR } 3,750$
2. $V_L = V_U + t \cdot D = 37,500 + 3,750 = \text{EUR } 41,250$, so $E = 41,250 - 15,000 = \text{EUR } 26,250$
3. $r_e = 10\% + (10\% - 5\%)(1 - 0.25) \times (15,000 / 26,250) = 12.143\%$
4. $\text{WACC} = (15/41.25) \times 5\% \times 0.75 + (26.25/41.25) \times 12.143\% = 1.36\% + 7.73\% = 9.091\%$

WACC falls 10% to 9.091%; value rises EUR 3,750, exactly the shield.

¹ Prop I with taxes: value rises by $t \cdot D$. Prop II with taxes: r_e still rises with D/E but more slowly, so WACC falls as leverage rises.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §4, Equations 5 and 6, Exhibits 5 and 6, and Example 3 "Gerhardt Corporation under Corporate Taxes", pp. 197–200.

Static trade-off theory: the optimum balances the tax shield against the PV of expected financial-distress costs

MM with taxes and no distress says borrow everything. Put a price on distress and the value curve turns over at some D^* — the optimal capital structure.

$$V_L = V_U + t \cdot D - \text{PV}(\text{costs of financial distress})$$

Debt D (USD m)	Tax shield $t \cdot D$	PV distress costs	V_L (USD m)	Δ vs V_U
0	0.0	0.0	45.0	—
5	1.5	0.1	46.4	+1.4
10	3.0	0.2	47.8	+2.8
15	4.5	0.5	49.0	+4.0
20 $\leftarrow D^*$	6.0	1.5	49.5	+4.5
25	7.5	6.0	46.5	+1.5
30	9.0	12.0	42.0	-3.0

Boulder Inc.: $V_U = \text{USD } 45\text{m}$, $t = 30\%$. $D^* = \text{USD } 20\text{m}$ maximises V_L at USD 49.5m; past that, distress costs grow faster than the shield. In practice D^* cannot be pinned down — managers set a range, say 30%–50% debt.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §5 "Optimal Capital Structure", Equation 7 and Exhibit 7, pp. 203–204; Boulder Inc. is Practice Problems 1–4, pp. 209–212.

Pecking order theory: information asymmetry pushes managers to internal funds first, then private debt, then public debt, and equity last

The ranking is by information content, not by cost. Managers prefer the source that reveals least — which is why an equity issue reads as bad news.

1. INTERNAL FUNDS

Retained earnings. No flotation cost, no disclosure, no signal.

First choice precisely because outsiders cannot read anything into it.

2. DEBT — PRIVATE BEFORE PUBLIC

If internal funds fall short, borrow — private debt ahead of public debt, because the lender already has the information.

Issuing debt is a mildly POSITIVE signal: you must be confident you can service it.

3. EQUITY — LAST

Last resort. An issue reads as management thinking the shares are dear — existing owners would not share a good thing.

Under pecking order there is no optimal structure: the mix just records past funding needs.

JENSEN (1986) — THE FREE-CASH-FLOW HYPOTHESIS

Debt disciplines. Contractual interest and principal force managers to use cash well and make empire-building acquisitions harder to fund, so greater use of debt can save agency costs of equity. A leveraged buyout of a cash-rich mature company exploits exactly that mechanism: replace discretionary cash with contractual debt service.

Source: CFA Institute, Curriculum Vol 3, Learning Module 6, §5 "Pecking Order Theory and Agency Costs", pp. 205–207; Myers & Majluf (1984), Jensen (1986).

SECTION 05

Cost of capital wrap

CAPM gives the cost of equity as the risk-free rate plus beta times the equity risk premium — only market risk is paid for

Only systematic risk earns a return; idiosyncratic risk is diversified away for free. Beta measures sensitivity to the market, and nothing else.

$$r_e = r_f + \beta \cdot (r_m - r_f) = r_f + \beta \cdot ERP$$

where: r_f = risk-free rate, usually the 10y government bond; β = levered equity beta; ERP = equity risk premium, historically 4%–6% in developed markets.

COST OF EQUITY FOR A TYPICAL LARGE-CAP INDUSTRIAL

Assumptions: r_f = 4.0% on the 10y government bond; ERP = 6.0% on long-run history; levered beta = 1.20, a cyclical manufacturer.

1. The stock's own risk premium: $\beta \times ERP = 1.20 \times 6.0\% = 7.20\%$
2. Add the risk-free rate: $r_e = 4.0\% + 7.20\%$
3. $r_e = 11.20\%$
4. Sensitivity: $\beta = 1.50$ gives 13.00%; $\beta = 0.80$ gives 8.80%. Every 0.10 of beta is 60bp.

$r_e = 11.2\%$ — and it carries most of the weight in any WACC.

¹ The Corporate Issuers text uses a cruder teaching anchor: the c.10% compound annual return on a broad equity index, 1928–2021.

Source: Cost-of-equity context: CFA Vol 3, Learning Module 6, pp. 178–179. The CAPM itself is developed in Vol 9, Portfolio Management, not in Corporate Issuers.

Unlever a comparable company's beta to isolate its business risk, then relever it at your target's own capital structure

An observed equity beta mixes business risk with the financial risk of that company's balance sheet. Unlevering separates the two; relevering rebuilds the one you want.

$$\beta_U = \beta_L / [1 + (1-t)D/E]$$

$$\beta_L = \beta_U [1 + (1-t)D/E]$$

where: β_U = unlevered, or asset, beta — business risk only; β_L = levered, or equity, beta; t = marginal tax rate. Assumes the debt beta is about zero.

UNLEVER THE PEER, RELEVER THE TARGET

The peer trades with a levered beta of 1.20 at D/E = 0.50 and a 25% tax rate. Your target runs at D/E = 1.00 — twice the gearing. What beta should you use for the target?

1. Unlever the peer: $\beta_U = 1.20 / [1 + (1 - 0.25) \times 0.50] = 1.20 / 1.375 = 0.873$
2. Relever at the target's D/E: $\beta_L = 0.873 \times [1 + (1 - 0.25) \times 1.00] = 0.873 \times 1.75$
3. $\beta_L = 1.527$
4. At $r_f = 4\%$ and ERP = 6%: the target's $r_e = 4\% + 1.527 \times 6\% = 13.16\%$

Beta 1.20 to 1.53, cost of equity +1.96pp, purely from the balance sheet.

Source: Hamada (1972), the market-standard beta adjustment. It is NOT part of CFA Level I — included because relevering a comparable beta is unavoidable in practice.

The after-tax cost of debt is the yield to maturity scaled by $(1 - t)$ — it is the yield that matters, never the coupon

The coupon is history — it fixes the cash interest on debt already issued. The yield is what the firm would pay to raise money today, which is what a discount rate needs.

$$\text{After-tax } r_d = \text{YTM} \cdot (1 - t)$$

where: Use the YTM on the issuer's own traded bonds; where nothing trades, match on rating, sector and maturity, or use recent comparable borrowings.

Rating / tenor (indicative)	Pre-tax YTM	After-tax, $t = 25\%$	After-tax, $t = 35\%$
A rated, 5y (mature industrial)	4.2%	3.15%	2.73%
BBB, 7y (capital-intensive)	5.4%	4.05%	3.51%
BB, 10y (leveraged buyout)	6.8%	5.10%	4.42%
B, 10y (speculative grade)	8.5%	6.38%	5.53%
CCC, 30y (distressed)	11.0%	8.25%	7.15%

WHY YIELD, NOT COUPON

A bond issued at a 5% coupon five years ago may yield 7% today. The 7% is the marginal cost — what the firm pays on new debt.

Source: Framework: CFA Institute, Curriculum Vol 3, Learning Module 6, §2, pp. 178–179. The yields shown are indicative market levels, not curriculum figures.

End to end: relever a peer beta, run CAPM, tax-effect the debt and combine — one discount rate out of five inputs

Five steps in order: pick the peer, unlever its beta, relever at the target D/E, run CAPM, then blend with the after-tax cost of debt.

AURELIA ENERGY CORP — A FULL WACC

A mid-cap independent power producer. Target structure 40% debt / 60% equity, so $D/V = 0.40$ and $D/E = 0.667$. Marginal tax rate 25%. $r_f = 4.0\%$, $ERP = 6.0\%$. A peer utility trades at a levered beta of 0.95 with $D/E = 0.80$. Aurelia's 10y unsecured bond yields 5.2%.

1. Unlever the peer: $\beta_U = 0.95 / [1 + (1 - 0.25)(0.80)] = 0.95 / 1.60 = 0.594$
2. Relever at Aurelia's $D/E = 0.667$: $\beta_L = 0.594 \times [1 + (1 - 0.25)(0.667)] = 0.594 \times 1.50 = 0.891$
3. CAPM cost of equity: $r_e = 4.0\% + 0.891 \times 6.0\% = 4.0\% + 5.35\% = 9.35\%$
4. After-tax cost of debt: $r_d \times (1 - t) = 5.2\% \times 0.75 = 3.90\%$
5. WACC = $0.40 \times 3.90\% + 0.60 \times 9.35\% = 1.56\% + 5.61\% = 7.17\%$

Aurelia WACC = 7.17%, for a project of average Aurelia risk.

¹ CAPM with a relevered peer beta is the default for listed comparables; for private companies a build-up with a size premium is more common.

Source: Illustrative company ("Aurelia Energy Corp"); synthesises the CFA Vol 3 LM 6 framework with the CAPM and the standard beta adjustment. Not a curriculum example.

Deck 9 — Corporate Governance, Capital Structure & Cost of Capital

- Five attributes separate the three legal forms: legal identity, owner-manager relationship, liability, taxation, access to financing. Corporations dominate because limited liability plus owner-manager separation opens capital markets.

- Leverage lifts ROE only while return on assets beats the cost of debt (CFA LM2 Example 1: 30% against 20%). It does not raise the average return — it widens the distribution.

- $WACC = w_d \cdot r_d \cdot (1 - t) + w_e \cdot r_e$. Market weights for the WACC; book values and leverage ratios for targets and covenants. The cost of debt is the YTM, never the coupon.

- Static trade-off adds distress costs and gives an optimum (Boulder: $D^* = 20$ maximises V_L at 49.5). Pecking order says there is none — internal funds, private debt, public debt, equity last.

- Debt is fixed, senior, finite and tax-deductible; equity is residual, discretionary and permanent. Debt is riskier for the issuer, equity for the investor. Both are true at once.

- Every principal-agent link creates agency costs: insufficient effort, wrong risk appetite, empire building, entrenchment, self-dealing. Seven categories of mechanism exist to contain them.

- MM without taxes: structure is irrelevant, cheaper debt exactly offset by dearer equity. With taxes: $V_L = V_U + tD$, WACC falls, and the extreme answer is 100% debt.

- Cost of equity from CAPM; the asset beta separates business risk from financial risk. Aurelia: unlever the peer, relever at target D/E, blend with after-tax debt, WACC = 7.17%.